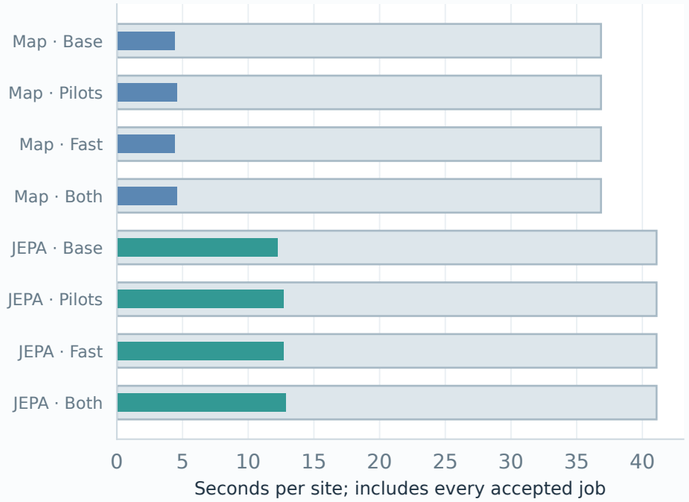
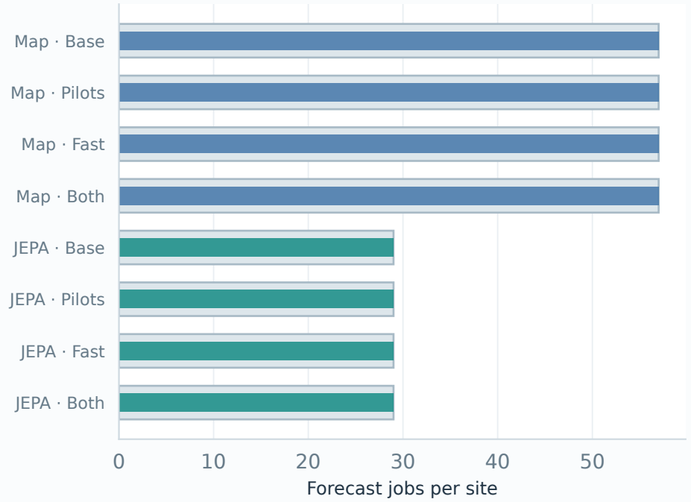
Compute, sensing and reporting costs remain visible

Means per 100-robot, 60-second site · actual-cost main rollouts · repetitions add no sites

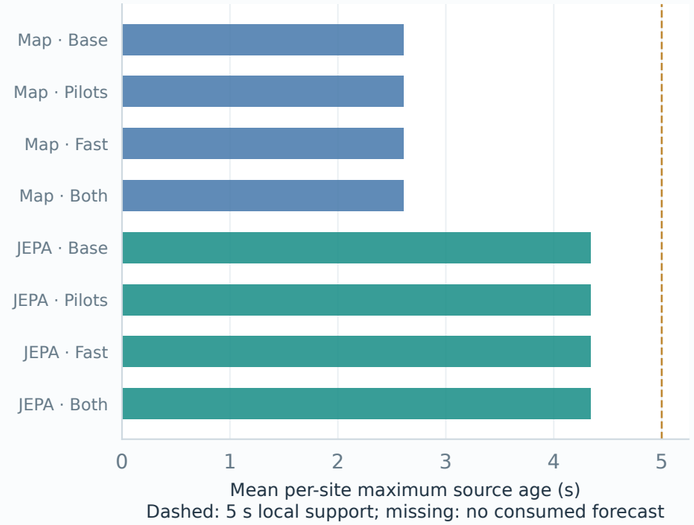
A Forecast compute is paid



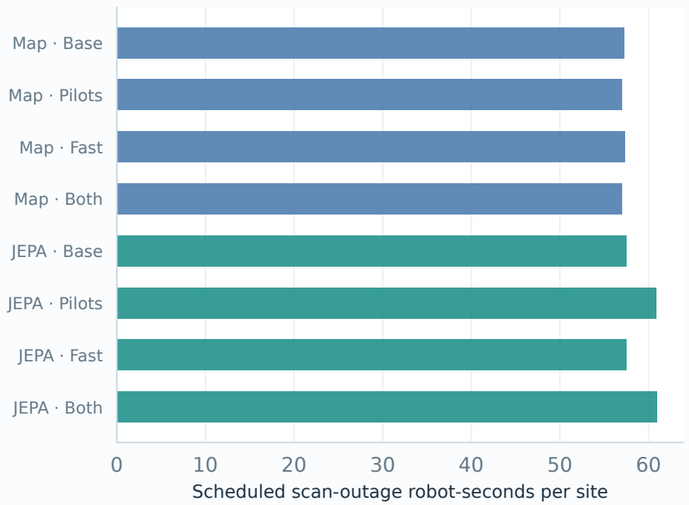
B Completed jobs drive planning



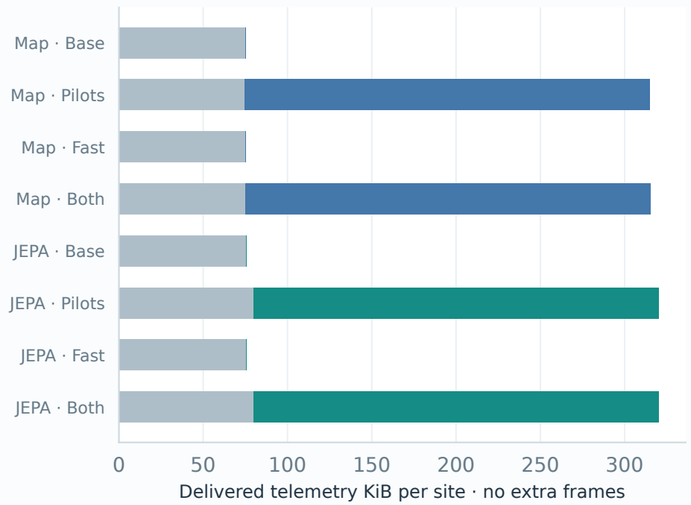
C Held forecasts keep aging



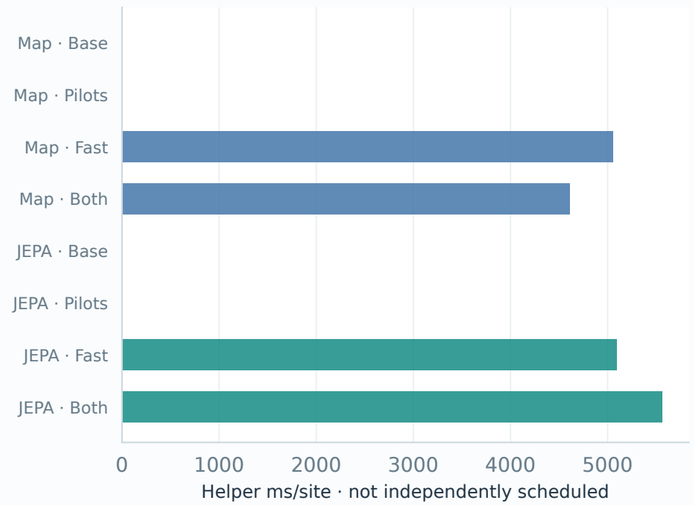
D Full scans interrupt the radio



E Existing control-report budget



F Local helper work is measured



A / B light bars: charged service / accepted jobs
A / B colored bars: measured compute / consumed jobs
E gray: delivered full-scan reservation
E colored: delivered 43-byte pilots

7,745,560 attempted original 600-byte control exchanges; 18,321.0 KiB attempted reports / 968,195.0 KiB existing reservation. 166,696 pilot rows retained.
Fast helper: 2,057,600 robot updates; largest measured call 7983.376 ms. No extra RF or model calls; pilot refresh <100 ms and capture <200 ms, with association guards.
Public pose/clock and zero intra-AP switch cost remain assumed. Forecast charge includes actual work and shared allowance. Unscheduled helper/observer CPU and modeled airtime do not establish complete-stack real-time operation.
Source / causal / paid / exact replay integrity: True. Simulation evidence; fixed neural weights, no hardware or production qualification.